

## Project Initialization and Planning Phase

|               |                                        |
|---------------|----------------------------------------|
| Date          | 29 June 2026                           |
| Team ID       |                                        |
| Project Title | <b>Credit Card Approval Prediction</b> |
| Maximum Marks | 5 Marks                                |

### Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

| Project Overview  |                                                                                                                                                                                                         |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Objective         | To develop an AI-powered Credit Card Approval Prediction System that predicts approval or rejection accurately and quickly using Machine Learning techniques.                                           |
| Scope             | The project covers data preprocessing, machine learning model training, prediction, web application development using Flask, and deployment on IBM Cloud for real-time credit card approval prediction. |
| Problem Statement |                                                                                                                                                                                                         |
| Description       | Traditional credit card approval processes are time-consuming, prone to human error, and may produce inconsistent decisions due to manual evaluation.                                                   |

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|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Impact                   | Delayed approvals, increased operational costs, inconsistent decisions, reduced customer satisfaction, and higher workload for financial institutions.                                                                                                                                                 |
| <b>Proposed Solution</b> |                                                                                                                                                                                                                                                                                                        |
| Approach                 | Build a Machine Learning classification model that analyzes applicant information and predicts whether the credit card application should be approved or rejected. The model is integrated into a Flask web application for real-time predictions.                                                     |
| Key Features             | <ul style="list-style-type: none"> <li>• Machine Learning-based approval prediction.</li> <li>• Real-time approval/rejection results.</li> <li>• User-friendly web interface.</li> <li>• Secure input validation.</li> <li>• Fast and accurate prediction.</li> <li>• IBM Cloud deployment.</li> </ul> |



## Resource Requirements

| Resource Type       | Description                           | Specification/Allocation       |
|---------------------|---------------------------------------|--------------------------------|
| <b>Hardware</b>     |                                       |                                |
| Computing Resources | CPU specifications                    | Intel Core i5/i7 or equivalent |
| Memory              | RAM specifications                    | 8 GB or higher                 |
| Storage             | Disk space for data, models, and logs | 20 GB free SSD (minimum)       |
| <b>Software</b>     |                                       |                                |

|                         |                      |                                                                                |
|-------------------------|----------------------|--------------------------------------------------------------------------------|
| Frameworks              | Python frameworks    | Flask                                                                          |
| Libraries               | Additional libraries | scikit-learn, pandas, numpy, matplotlib, seaborn                               |
| Development Environment | IDE                  | Jupyter Notebook, Visual Studio Code or PyCharm                                |
| <b>Data</b>             |                      |                                                                                |
| Data                    | Source, size, format | Credit Card Approval Dataset (CSV) from Kaggle/UCI Machine Learning Repository |